INDUSTRIAL PREDICTIVE MAINTENANCE & FAILURE ANALYSIS

Executive Summary | Microsoft Excel portfolio project | AI4I 2020 Predictive Maintenance Dataset

1. EXECUTIVE SUMMARY

This project analyzes machine failure patterns in the AI4I 2020 Predictive Maintenance Dataset using Microsoft Excel. Its purpose is to identify operational conditions associated with higher observed failure rates and translate them into practical preventive maintenance monitoring priorities. The work is descriptive and does not establish causal relationships.

2. BUSINESS PROBLEM

Unexpected machine failures can disrupt production, increase maintenance costs, and reduce asset availability. Identifying recurring patterns helps maintenance teams focus inspection capacity on higher-risk operating conditions and improve preventive intervention planning.

3. DATASET OVERVIEW

Metric	Verified Detail
Dataset	AI4I 2020 Predictive Maintenance Dataset
Total Records	10,000
Machine Failures	339
Overall Failure Rate	3.39%

The 10,000-record dataset contains machine and product identifiers, product type, air and process temperatures, rotational speed, torque, tool wear, an overall machine-failure indicator, and five failure-category indicators. These variables support descriptive comparisons of failure patterns across operational ranges.

4. DATA QUALITY

Quality checks confirmed 10,000 records, no missing values in the analyzed variables, no duplicate UDI values, and no duplicate Product ID values. Numeric, text, and failure-indicator fields were validated before analysis, strengthening confidence in the descriptive outputs.

5. METHODOLOGY

Analytical Step	Approach	Purpose	Output
Excel-based descriptive analysis	Descriptive statistics; failure vs. non-failure comparison; failure-category analysis; and banded torque, tool wear, air-temperature, and process-temperature analysis.	Identify operational ranges associated with higher observed failure rates.	Dashboard, analysis sheets, Key Insights, and executive reporting.

6. KEY FINDINGS

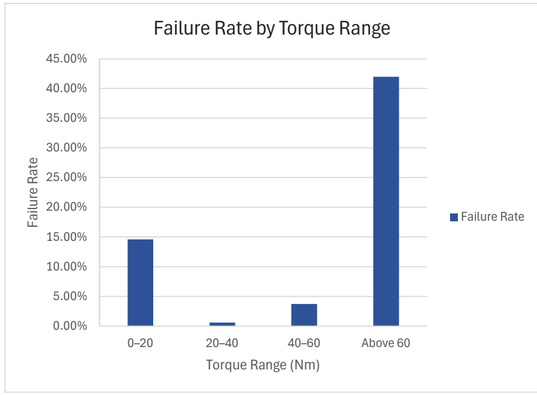
Analysis Area	Verified Finding	Business Interpretation	Recommended Action
Overall Machine Failure	10,000 total records; 339 machine failures; 3.39% overall failure rate.	Failures represent a small but operationally meaningful share of observed records. This is an observed dataset outcome, not evidence of any single cause.	Use the 3.39% rate as a baseline for maintenance review and prioritize inspection using the higher-risk operating patterns below.
Torque Analysis	Above 60 Nm: 41.95% failure rate; 20-40 Nm: 0.59% failure rate.	Failure rates are substantially higher in the above-60 Nm band than in the 20-40 Nm band. This is an association in the observed records and does not prove that torque directly causes failure.	Prioritize inspection of high-torque operating conditions and review machines operating above 60 Nm within preventive maintenance planning.
Tool Wear Analysis	201-253 minutes: 15.49% failure rate; 151-200 minutes: 2.90% failure rate.	The highest provided tool-wear band is associated with a higher failure rate than the preceding band. The relationship is observational, not causal.	Monitor machines with higher tool wear and prioritize inspection as tool wear approaches the 201-253 minute range.
Air Temperature Analysis	303-305 K: 6.24% failure rate; 297-299 K: 2.05% failure rate.	The higher air-temperature band shows a higher observed failure rate than the lower comparison band. Temperature may be a useful monitoring signal, but causation is not established.	Track air-temperature patterns and flag sustained operation in the 303-305 K band for maintenance review.
Process Temperature Analysis	311-313 K: 3.88% failure rate; 305-307 K: 2.33% failure rate.	The 311-313 K band has a higher observed failure rate than the 305-307 K band. This comparison indicates association only.	Include process-temperature patterns in preventive inspection priorities and review repeated operation in the 311-313 K band.

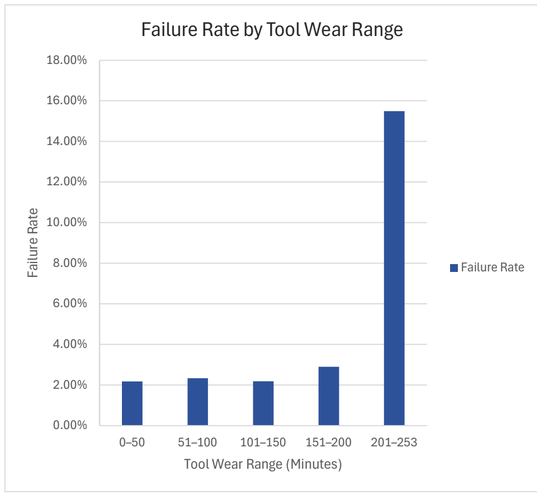
7. BUSINESS RECOMMENDATIONS

Priority	Recommended Monitoring Action	Operational Rationale	Governance
Preventive maintenance inspection	Prioritize inspection of high-torque operating conditions; monitor machines with higher tool wear; track air and process temperature patterns; combine these indicators when setting inspection priorities.	Direct maintenance attention toward conditions associated with higher observed failure rates while retaining engineering judgment and existing safety procedures.	Validate monitoring rules on real plant data before using them for automated maintenance decisions.

8. LIMITATIONS

AI4I 2020 is a synthetic predictive-maintenance dataset and may not reflect the equipment mix, operating environment, maintenance history, or failure mechanisms of a specific facility. This analysis identifies associations between operating bands and failure rates; it does not prove causation. Existing workbook bands should not be





9. CONCLUSION

The project demonstrates an end-to-end Excel analytics workflow covering data-quality validation, descriptive analysis, comparative failure assessment, dashboard design, and business communication. Its value lies in converting verified patterns into transparent inspection priorities while clearly separating observed evidence